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Roll No

CM-601 (GS)
B.E. VI Semester Examination, May 2020
Grading System (GS)
Process Equipment Design - I
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Give the procedure of the design of saddle supports for a horizontal vessel. Explain all term used.
2. As a design engineer, which are the various pressure tests carried out for the design of pressure vessel, explain them in brief?
3. Explain, what do you understand by class 1, class 2 and class 3 types of pressure vessels.
4. A pressure vessel with outer radius to inner radius ratio of 1.75 is to be used for high pressure service. The yield stress of the material is 5000kg/cm^2 . The factor of safety is 1.75.
 - i) Find the maximum pressure the vessel can withstand.
 - ii) If the vessel is used for an internal pressure of 50 percent maximum pressure, find the resultant stress that will be induced in the vessel.
5. Discuss how wind load is taken care of in the design of tall vertical vessel. What are skirt supports? Discuss the method for calculating the thickness of a skirt support.

OR

Explain different types of head and closures of pressure vessel.

6. A thick wall pressure vessel of steel having an inside diameter of 200mm and outside diameter of 400mm is subjected to an internal pressure of 85mN/m^2 . Determine the maximum induced stress according to:
 - i) Maximum principle stress theory
 - ii) Maximum Shear stress theory
 - iii) Maximum strain theoryYield point of steel = 450mN/m^2 .
Factor of safety = 1.5
7. Write notes on any three
 - a) Design stress and factor of safety
 - b) Different types of flange faces
 - c) Power required for Agitation
 - d) Different types of floating roots
8. Write basic steps that we go through in the process of fabrication or welding up any metal fabricated object.
